



EDGEPULSE

Edge-AI Predictive Maintenance and Health Forecasting for Industrial Heavy Machinery

Team-

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Category: AI at the Edge Solutions for Industrial Heavy Machinery

Project Status: Proof of Concept (Work in Progress)



INTRODUCTION & PROJECT OVERVIEW

Industrial heavy machinery often experiences unexpected failures that lead to production delays, expensive repairs, and operational downtime.

Traditional maintenance approaches either:

- Reactive Maintenance (after failure)
- Preventive Maintenance (fixed schedule)

Both approaches increase cost and reduce efficiency.

EdgePulse proposes an Edge-AI based predictive maintenance system that continuously monitors machine vibration data and predicts equipment health before failures occur.

PROBLEM STATEMENT + MARKET OPPORTUNITY

Problem:

Industrial rotating machines such as pumps, excavators, and heavy industrial systems generate vibration patterns before failure occurs. Existing monitoring approaches often rely on centralized cloud infrastructure and delayed decision-making.

Current Challenges:

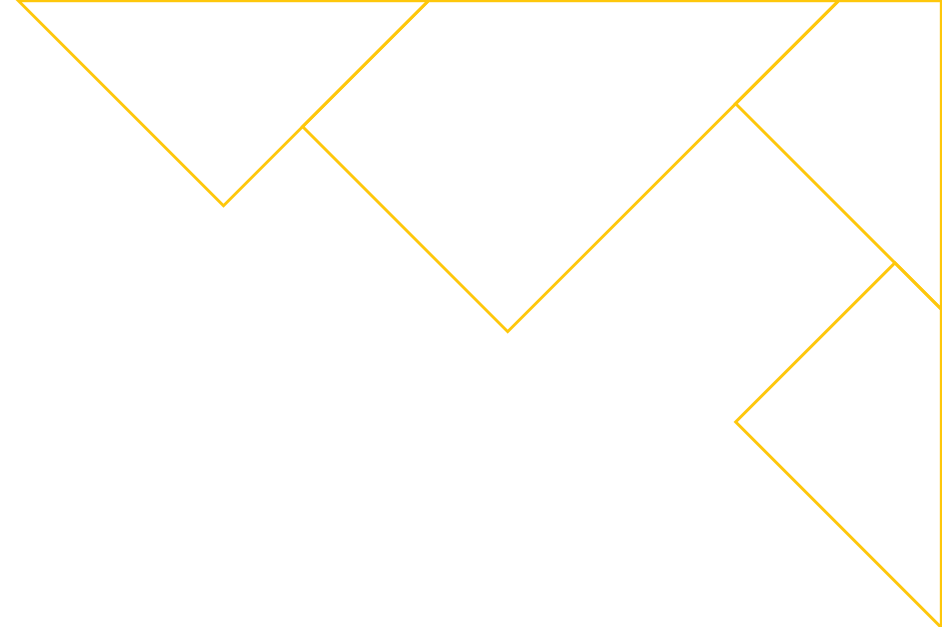
- Unexpected downtime
- High maintenance costs
- Network dependency
- Excessive raw data transfer
- Delayed fault detection

Opportunity:

Predictive maintenance and Edge AI can reduce downtime and improve machine reliability while enabling faster local decisions.

Enable low-latency and cost-efficient industrial monitoring



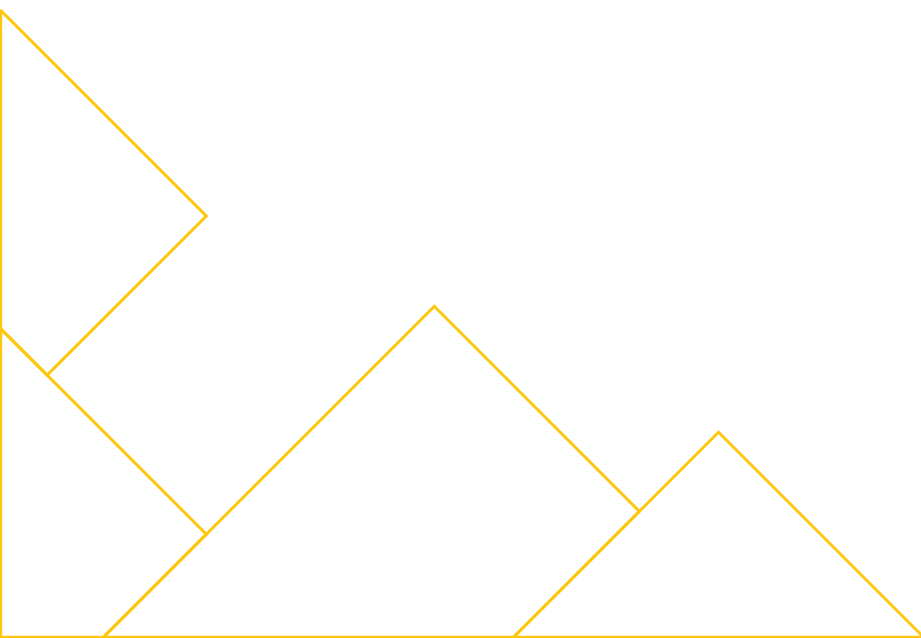


OBJECTIVE AND APPROACH

Objective:

Develop an Edge-AI system capable of predicting machine degradation stages locally using vibration data.

Approach:

1. Collect industrial vibration dataset
 2. Preprocess sensor signals
 3. Extract useful statistical features
 4. Train predictive ML model
 5. Prepare lightweight edge inference pipeline
 6. Generate maintenance alerts
 7. Transmit only summarized insights to dashboard
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SOLUTION OVERVIEW (ARCHITECTURE)

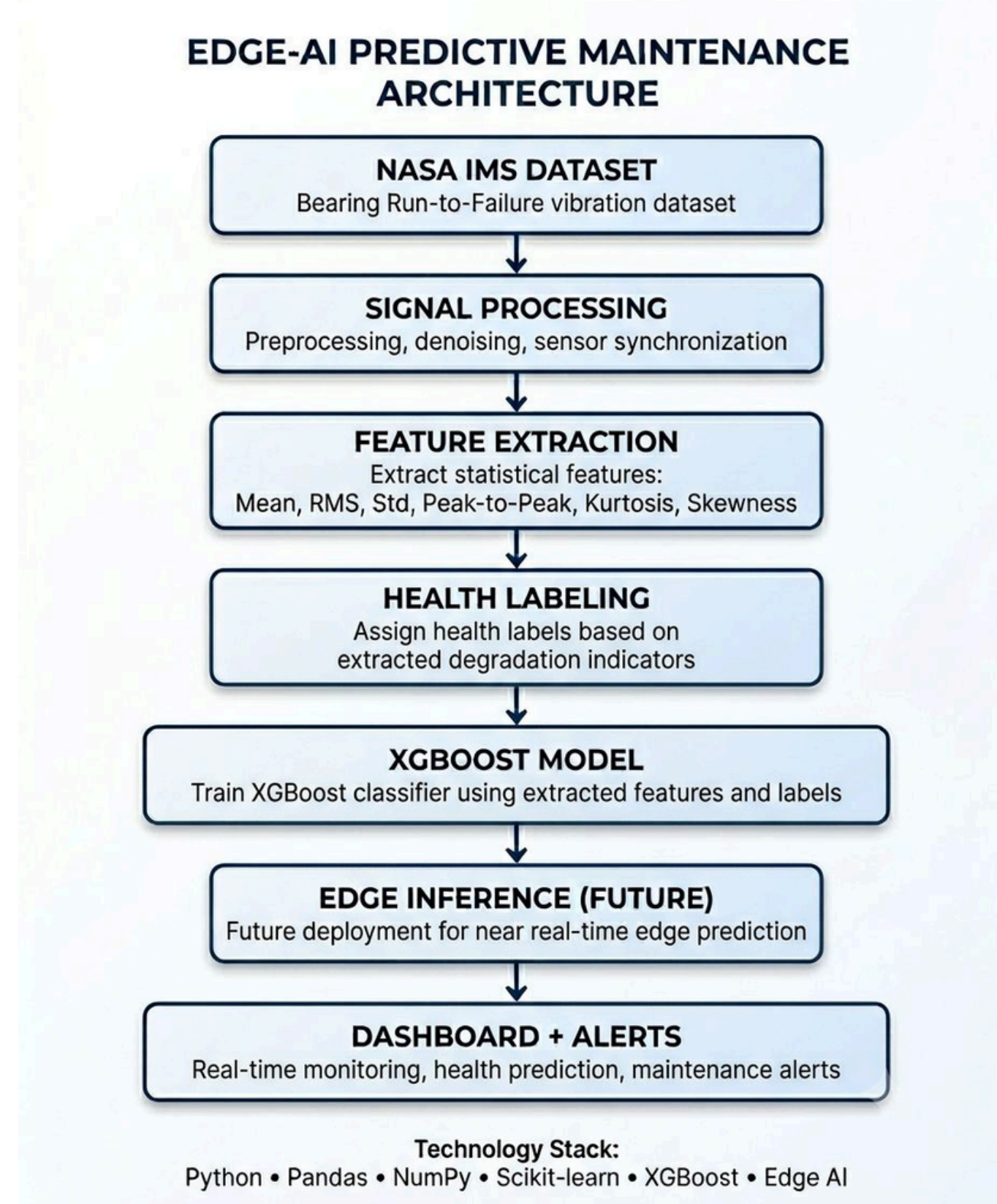


Figure: EdgePulse System Architecture

DATASET AND TECHNICAL IMPLEMENTATION

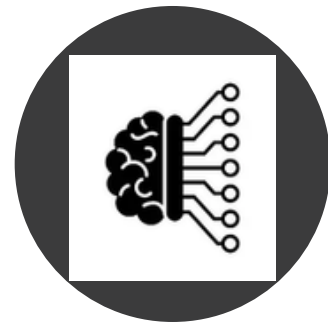
Development Workflow:

Dataset → Training → Validation → Edge Deployment → Dashboard



Dataset:

NASA IMS Bearing Dataset



Machine Type:

Industrial Rotating Machinery



Technologies Used:

- Python
- NumPy
- Pandas
- Scikit-learn
- XGBoost
- ONNX Runtime
- FastAPI
- Docker
- Git & GitHub
- Jupyter Notebook

NOVELTY AND DIFFERENTIATION

What makes EdgePulse different:

Traditional Systems:

- Cloud dependent
- High latency
- Continuous raw data transfer

EdgePulse:

- Edge-first inference
- Low bandwidth usage
- Faster decisions
- Cloud synchronization using summarized data
- Explainable predictions using SHAP
- Edge inference reduces dependency on continuous cloud connectivity

CHALLENGES FACED

Challenges :

- Large dataset extraction and preprocessing
- Understanding degradation signals
- Managing edge deployment constraints
- Designing lightweight inference
- Creating scalable architecture

Solutions:

- Structured data pipeline
- Model optimization
- Modular project design
- Local processing approach

CURRENT RESULTS AND PROGRESS

Current Status:

Proof-of-Concept (PoC) — Development in Progress

Completed:

- ✓ Dataset downloaded and validated
- ✓ Data extraction completed
- ✓ Project structure created
- ✓ Data exploration completed
- ✓ System architecture finalized
- ✓ Initial data processing pipeline prepared

In Progress:

- Feature engineering
- Model training
- ONNX deployment
- Dashboard integration

EXPECTED BENEFITS AND IMPACT

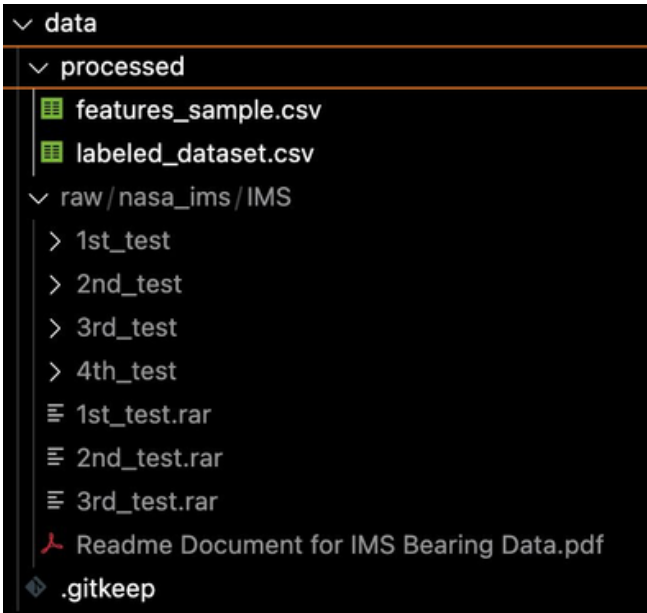
Expected Outcomes:

- Reduced downtime through earlier fault detection
- Faster maintenance decisions
- Lower cloud dependency
- Reduced unnecessary inspections
- Better equipment reliability

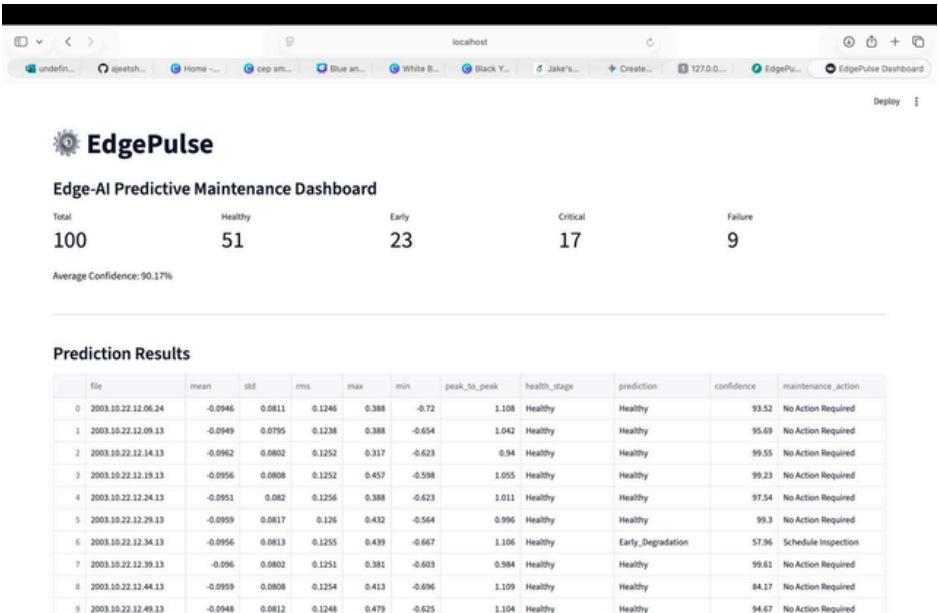
Long-Term Impact:

Move industries from reactive maintenance toward predictive operations.

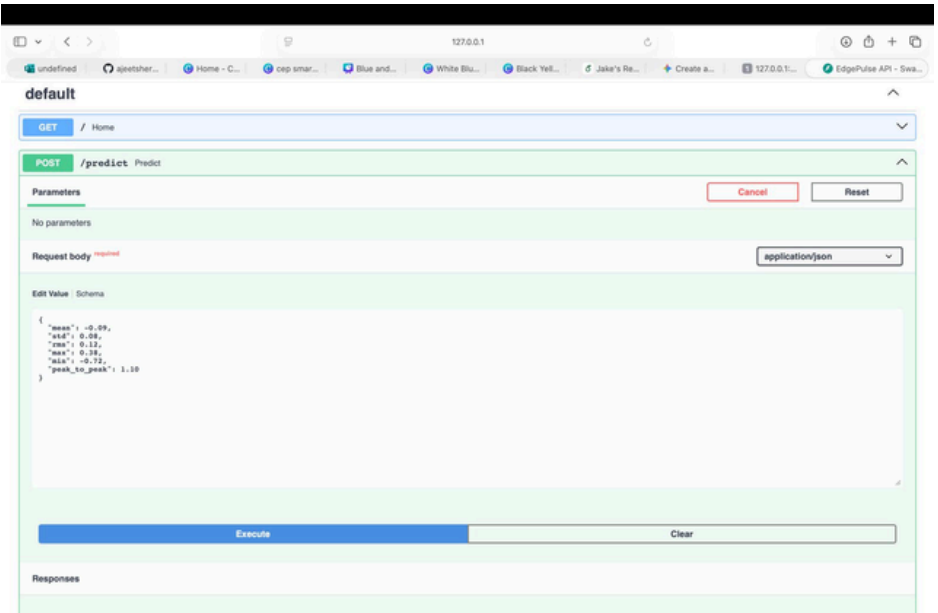
Demonstration & System Validation



- Dataset Exploration & Feature Engineering



- Prediction Dashboard with Monitoring



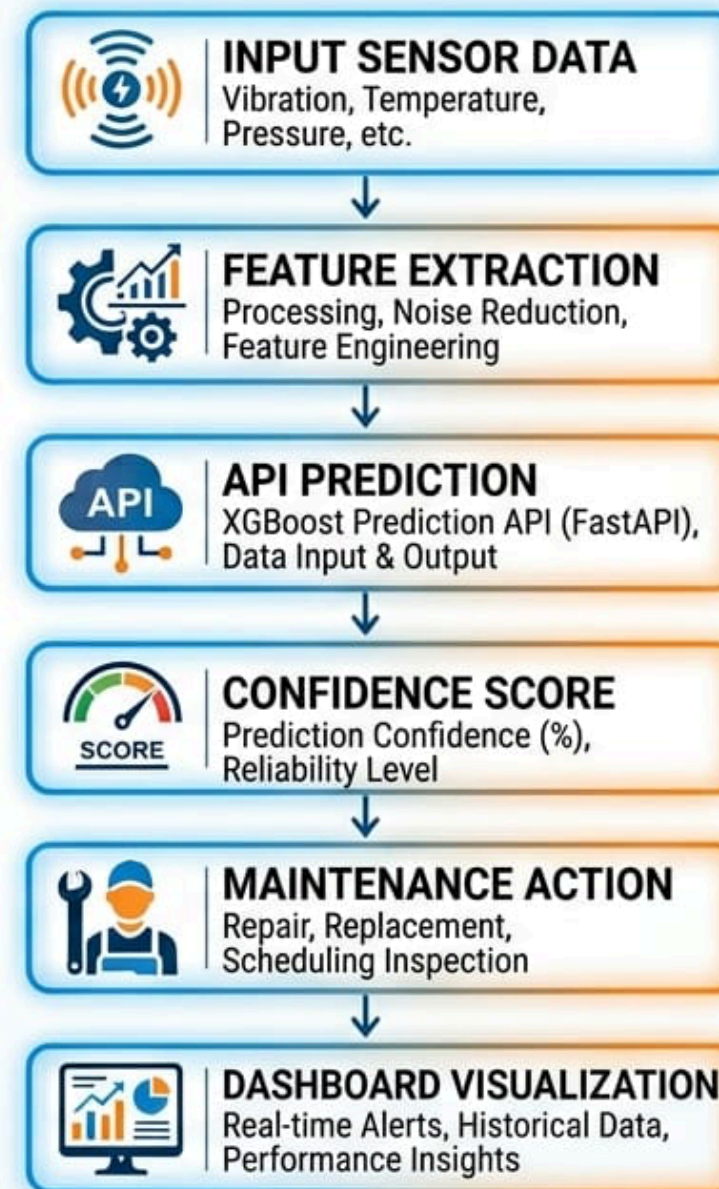
- FastAPI Prediction API Testing



- Health Prediction & Maintenance Recommendation

Predictive Maintenance Workflow

DEMONSTRATION WORKFLOW: PREDICTIVE MAINTENANCE



Future Enhancements & Project Plan

Future Scope:

- Real-time sensor integration
- Digital Twin support
- Mobile dashboard
- Multi-machine deployment
- Edge hardware optimization
- Model compression and optimization for edge devices

THANK
YOU

